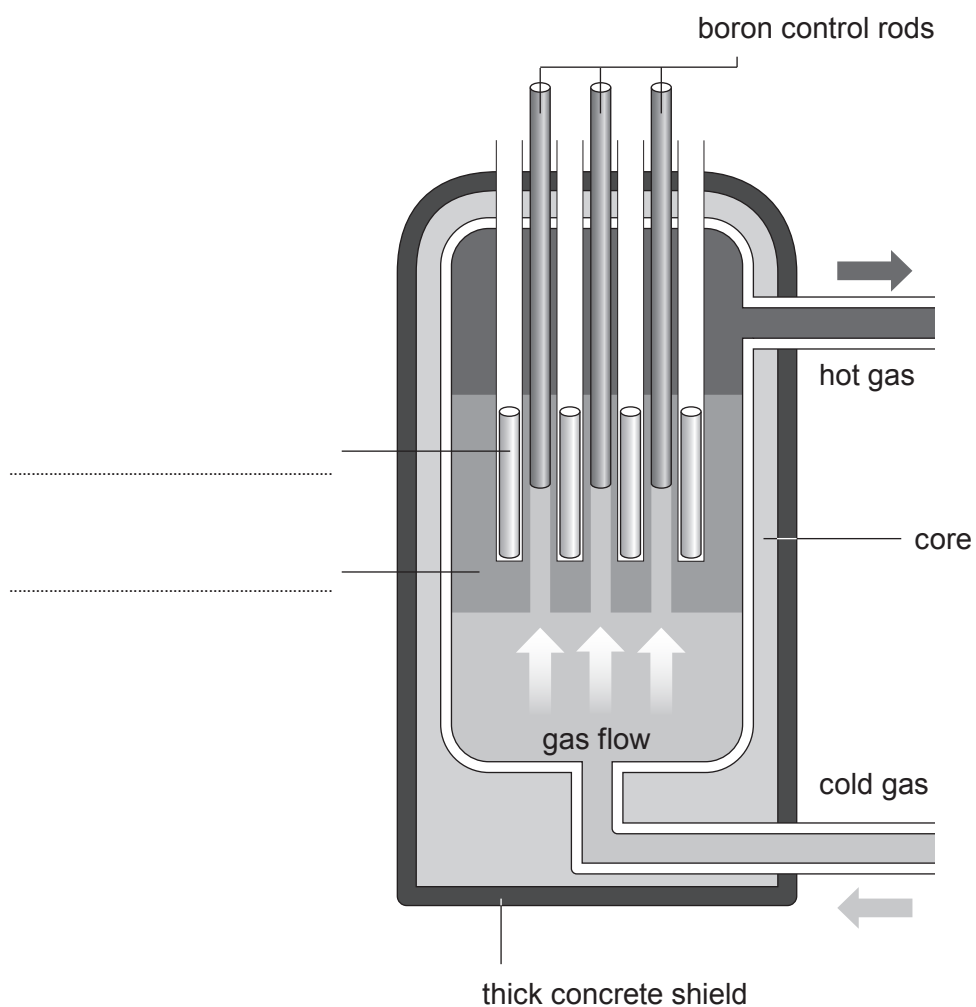


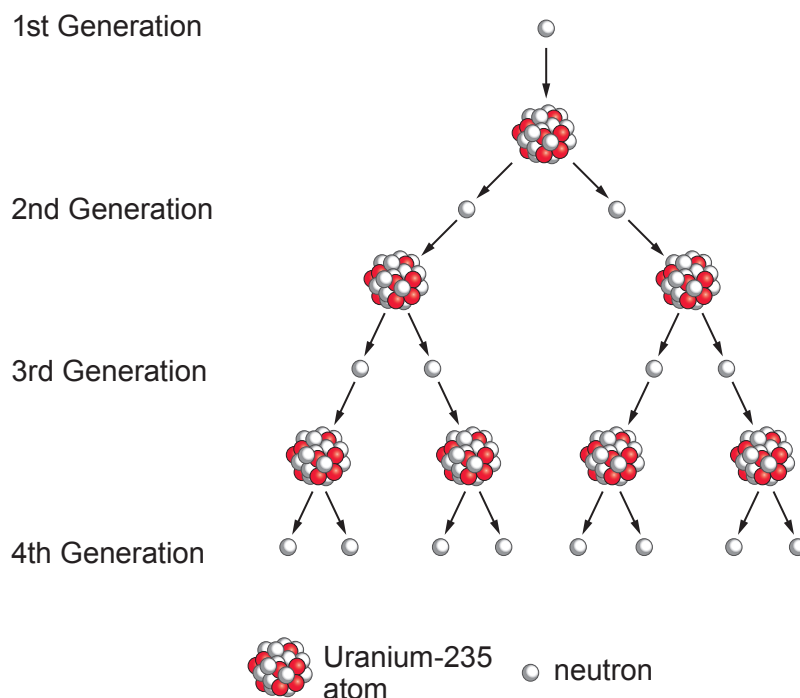
4. Nuclear fission occurs when a nucleus of uranium is split after absorbing a neutron. It is the process used in a nuclear reactor like the one shown in the diagram below.



(a) Complete the labelling of the diagram.

[2]

- (b) The diagram below shows an uncontrolled chain reaction of the decay of uranium.



- (i) Calculate the number of 5th generation neutrons. [1]
- (ii) Explain how the number of neutrons in your answer above would change if the control rods are lowered further into the reactor. [2]

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- (iii) State the danger of an uncontrolled chain reaction in a nuclear reactor. [1]

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